
Lamprop manual

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1 Introduction

The purpose of this program is to calculate some properties of fiber-reinforced composite laminates. It calculates:

- engineering properties like E_x , E_y and G_{xy}
- thermal properties like α_x and α_y
- physical properties like density and laminate thickness
- stiffness and compliance matrices (ABD and abd)

This program can *not* calculate the strength of composite laminates; because there are many different failure modes, strengths of composite laminates cannot readily be calculated from the strengths of the separate materials that form the laminate. These strengths really have to be determined from tests. However, the author has found [Hart-Smith \[1992\]](#) useful for initial estimation of the strengths of multi-layer laminates.

The generally hard to obtain transverse fiber properties were replaced with properties derived from the matrix.

The books from [Hyer \[1998\]](#) and [Tsai \[1992\]](#) and the report from [Nettles \[1994\]](#) were instrumental in writing the code. More recently, the books from [Bower \[2010\]](#) and [Barbero \[2008, 2018\]](#) have proved crucial for the author's understanding of the subject matter and thus improvements in the code. The article [Giner et al. \[2014\]](#) provided a refinement in the estimation of E_2 .

In this manual, UNIX-like systems like Linux, FreeBSD, Open BSD, NetBSD and macOS will be referred to as "POSIX" for brevity.

2 Requirements

To build this program you will need a C compiler that supports the C11 standard. The build has been tested with `clang` version 19 and `gcc` version 15.

Using the `make` program is optional but recommended. Both GNU `make` and BSD `make` have been shown to work.

2.1 POSIX

On a POSIX system, the program can be built by simply invoking `make`.

2.2 Ms-windows

The only tested build method on ms-windows is using `w64devkit`. Open `w64devkit`, change to the directory where you have stored the source, and issue the command `make -f Makefile.win32`.

3 Using the program

The program does not *require* installation to run. Only the binary is required.

For convenience you can copy the binary to a location mentioned in the environment variable `PATH` (POSIX) or `Path` (ms-windows).

`lamprop` processes input files with laminate descriptions. These files are written in plain text, and have the extension `lam`. The format of these files is documented in the next chapter.

The `lamprop` program runs on the command line.

3.1 POSIX

Assuming that `lamprop` is installed in your `PATH`, simply call it as `lamprop` from your shell, X11 or Wayland terminal emulator.

If the location is not in your `PATH`, call the program with its location:
`~/local/bin/lamprop -h`.

3.2 Ms-windows

To run `lamprop.exe`, open the folder where the `lam` file you want to process resides in the windows explorer. Right-click in that folder, and select “Open in Terminal”.

To run `lamprop.exe`, you have to specify the complete path where the program is located, unless it is stored in a location that is located in the `Path` environment variable. When there are spaces in the path, the whole path has to be enclosed in double quotation marks, and it has to be preceded by an ampersand, for example:

```
& "C:\users\John Doe\lamprop\lamprop.exe" -h.
```

4 The lamprop file format

The file format is very simple. Functional lines have either `f`, `r`, `t`, `m`, `l`, `c` or `s` as the first non white space character. This character must immediately be followed by a colon `:`. All other lines are seen as comments and disregarded.

This program assumes specific metric units. The units used below are important because the program internally calculates the thickness of layers (in mm) based on the volume fractions and densities of the fibers and resins.

The `f` : -line contains a definition of a fiber. The parser converts this into an instance of a `Fiber` object. The line must contain the following values, separated by white space:

E_1 Young's modulus in the fiber direction in MPa.
 ν_{12} Poisson's constant (dimensionless).
 α_1 Coefficient of Thermal Expansion in the fiber direction in K^{-1} .
 ρ Density of the fiber in g/cm^3 .
name The identifier for the resin. This should be unique in all the files read. Contrary to the previous values, this may contain whitespace.

Below an example for standard e-glass fiber.

```
73000 0.33 5.3e-6 2.60 e-glass
```

Usually, E_1 and other properties in the fibre length direction are easily obtained from a fiber supplier. Previous versions of this program also required the Young's modulus perpendicular to the fiber to calculate transverse properties of the lamina. Since this value is generally not given in the manufacturer documentation, it has been replaced by a value calculated with the Halpin-Tsai formula from [Barbero \[2018\]](#), p. 117]. Previously the modulus of the matrix multiplied by a factor was used, according to [Tsai \[1992\]](#). However, the author has found that the factor provided by Tsai overestimates E_2 . So the factor ξ was reduced from 2 to 1.5 conform [Giner et al. \[2014\]](#)

The `r` : -line contains a definition of a resin. Like with the fibers, this becomes an

instance of a `Resin` object in the code. The resin line must contain the following values, separated by white space.

E Young's modulus in MPa.
 ν Poisson's constant (dimensionless).
 α Coefficient of Thermal Expansion in K^{-1} .
 ρ Density of the resin in g/cm^3 .
name The identifier for the resin. This should be unique in all the files read. Contrary to the previous values, this may contain whitespace.

An example of a generic thermoset resin is shown below.

```
3800 0.36 40e-6 1.165 generic
```

The `t` : line starts a new laminate. It only contains the name which identifies the laminate. This name must be unique within the current input files. It may contain spaces.

The `m` : line chooses a resin for the laminate. It must appear after a `t` : line, and before the `l` : lines. It must contain the following values, separated by white space:

vf The fiber volume fraction. This should be a number between 0 and 1 or between 1 up to and including 100. In the latter case it is interpreted as a percentage.
name The name of the resin to use. This must have been previously declared with an `r` : -line.

The `l` : line defines a single layer (lamina) in the laminate. It must be preceded by a `t` : and a `m` : line. It must contain the following values, separated by white space (optional items in brackets):

weight The area weight in g/m^2 of the dry fibers.
angle The angle upwards from the x-axis under which the fibers are oriented.
(vf) Optionally the layer can have a different fiber volume fraction.

name The name of the fiber used in this layer. This fiber must have been declared previously with an `f :` line.

The last line in a laminate definition can be an `s :` line, which stands for “symmetry”. This signifies that all the lamina before it are to be added again in reverse order, making a symmetric laminate stack. An `s :` line in any other position is an error.

An example is given below.

```
Fiber definition
  E1      v12  alpha1  rho  naam
f: 233000 0.2   -0.54e-6 1.76 Hyer's carbon fiber

Matrix definition
  Em      v      alpha  rho name
r: 4620 0.36 41.4e-6 1.1 Hyer's resin

t: [0/90]s laminate
This is a standard symmetric cross-ply laminate. It has fine extensional
moduli in the fiber directions, but a very low shear modulus.
m: 0.5 Hyer's resin
l: 100 0 Hyer's carbon fiber
l: 100 90 Hyer's carbon fiber
s:
```

The program has the following storage limits:

- 1000 resins
- 1000 fibers
- 25000 lamina
- 100 laminates

By using the `-i` option, you can see how many of that space has been used.

The author has personally used laminates with up to 250 layers. Calculating the properties of that laminate took approximately 0.005 s on a machine with an Intel Core2 Q9300 running FreeBSD.

Interspersed between the `l :` lines (and before the `s :` line) there can be `c :` lines. These are comments about the lay-up, that will be inserted into the output. Their use is to signify that the `l :` lines following a comment are part of a subassembly such as a woven fabric or a non-crimp fabric.

For example:

```
  E1 v      12 alpha      1 rho      name
f: 70000 0.33 5e-6 2.54 e-glas

  Em v op      name
r: 3600 0.36 60e-6 1.145 atlac-430

t: atlac430-combi
```

```
m: 0.22 atlac-430
c: CSM 450
l: 150 0 e-glas
l: 150 60 e-glas
l: 150 -60 e-glas
c: UNIE640
l: 600 0 0.50 e-glas
l: 40 90 0.50 e-glas
c: -- symmetric --
s:
```

This will yield the output shown in Table 4.1 on page 8.

4.1 Material data

Over the years, the author has gathered a lot of data for different fibers from data sheets provided by the manufacturers. Data for different fibers is given in Table 4.2. In case the ν_{12} is not known for a carbon fiber, it is estimated at 0.2. Similarly, if the α_1 is not known, it is estimated at $-0.12 \times 10^{-6} \text{ K}^{-1}$. For glass fibers, ν_{12} is estimated 0.33 unless known and α_1 is estimated $5 \times 10^{-6} \text{ K}^{-1}$ unless known.

Several resins are shown in Table 4.3. For resins, ν is estimated 0.36 unless known and α is estimated $40 \times 10^{-6} \text{ K}^{-1}$ unless known.

4.2 Using the command-line program

The command `lamprop -h` produces the following overview of the options.

```
usage: lamprop [-h] [-i | -l | -H] [-e] [-m] [-f] [-L | -v]
               [file ...]
```

Calculate the elastic properties of a fibrous composite laminate. See the manual (lamprop-manual.pdf) for more in-depth information.

positional arguments:

file one or more files to process

options:

```
-h, --help          show this help message and exit
-i, --info          show information about source file (the default is not to)
-l, --latex         generate LaTeX output (the default is plain text)
-H, --html          generate HTML output
-e, --eng           output only the engineering properties
-m, --mat           output only the ABD matrix and stiffness tensor
-f, --fea           output only material data for FEA
-L, --license       print the license
-v, --version       show program's version number and exit
```

Running `lamprop` from the command line produces text output by default. Output in $\text{\LaTeX}$ or HTML format can be requested with the appropriate arguments. As of 4.0, RTF output (for inclusion in word processor documents) has been removed. Since most word processors can read HTML, use that instead.

Table 4.1: properties of atlac430-combi

generated by lamprop version 2023.10.10

Laminate stacking					
Layer	Weight [g/m ²]	Angle [°]	vf [%]	Fiber type	
CSM 450					
1	150	0	22	e-glas	
2	150	60	22	e-glas	
3	150	-60	22	e-glas	
UNIE640					
4	600	0	50	e-glas	
5	40	90	50	e-glas	
– symmetric –					
UNIE640					
6	40	90	50	e-glas	
7	600	0	50	e-glas	
CSM 450					
8	150	-60	22	e-glas	
9	150	60	22	e-glas	
10	150	0	22	e-glas	
Physical properties					
Property	Value	Dimension			
v _f	32.8	%			
w _f	52	%			
thickness	2.62	mm			
density	1.6	g/cm ³			
weight	4195	g/m ²			
resin	2015	g/m ²			
Engineering properties					
In-plane			3D stiffness tensor		
E _x	19544	MPa	E _x	19549	MPa
E _y	10958	MPa	E _y	10968	MPa
E _z	7004	MPa	E _z	8344	MPa
G _{xy}	3758	MPa	G _{xy}	3759	MPa
G _{xz}	2538	MPa	G _{xz}	2736	MPa
G _{yz}	2437	MPa	G _{yz}	2632	MPa
ν _{xy}	0.3207	-	ν _{xy}	0.3213	-
ν _{yx}	0.1798	-	ν _{xz}	0.3748	-
α _x	9.854e-06	K ⁻¹	ν _{yz}	0.4463	-
α _y	1.924e-05	K ⁻¹			

Table 4.2: fibers

Name	E ₁ [MPa]	ν ₁₂ [-]	α ₁ [K ⁻¹]	ρ [g/cm ³]	Type
Tenax HTA	238000	0.20	-0.1e-6	1.76	carbon
Tenax HTS	240000	0.20	-0.1e-6	1.77	carbon
Tenax STS40	240000	0.20	-0.12e-6	1.78	carbon
Toracya T300	230000	0.27	-0.41e-6	1.76	carbon
Torayca T700SC	230000	0.27	-0.38e-6	1.80	carbon
pyrofil TR30S	235000	0.20	-0.5e-6	1.79	carbon
sigrafil CT24-5.0-270/E100	270000	0.25	-0.12e-6	1.79	carbon
K63712	640000	0.234	-1.47e-6	2.12	carbon
K63A12	790000	0.23	-1.2e-6	2.15	carbon
Torayca T800S	294000	0.27	-0.60e-6	1.76	carbon
K13C2U	900000	0.234	-1.47e-6	2.20	carbon
M35J	339000	0.27	-0.73e-6	1.75	carbon
M46J	436000	0.234	-0.9e-6	1.84	carbon
PX35UD	242000	0.27	-0.6e-6	1.81	carbon
Granoc XN-80-60S	780000	0.27	-1.5e-6	2.17	carbon
Granoc XN-90-60S	860000	0.27	-1.5e-6	2.19	carbon
e-glass	73000	0.33	5.3e-6	2.60	glass
ecr-glass	81000	0.33	5e-6	2.62	glass

Table 4.3: Resins

Name	E [MPa]	ν [-]	α [K ⁻¹]	ρ [g/cm ³]	Type
Epikote EPRO4908	2900	0.25	40e-6	1.15	epoxy
Palatal P4-01	4300	0.36	40e-6	1.19	polyester
Synolite 2155-N-1	4000	0.36	40e-6	1.22	polyester
Distitron 3501LS1	4100	0.36	40e-6	1.2	polyester
Synolite 1967-G-6	3800	0.36	40e-6	1.165	DCPD
atlac 430	3600	0.36	55e-6	1.145	vinylester

4.3 Meaning of the ABD, H and C matrices

The stiffness or ABD matrix are what converts in-plane strains into forces and the other way around, see Table 4.5. It is a 6×6 matrix that can be divided into three 3×3 matrices; A, B and D. The expansions below reveal the symmetries in this matrix.

$$ABD = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix}$$

The units of the parts of the ABD matrix are as follows (where i and j are 1, 2 or 6): A_{ij} is in N/mm. B_{ij} is in Nmm/mm = N. D_{ij} is in N mm.

The stress resultants N are units of force per unit of width (N/mm). Moment resultants m are in units of torque per unit of width (Nmm/mm = N). Both strains ϵ and κ are dimensionless. The ABD and H matrix equations in Table 4.4 basically show the behavior of a square piece of laminate small enough that the stress and strain resultants can be considered constant over its dimensions.

If we ignore thermal effects, the most general form of linear elasticity is:

$$\sigma_{ij} = C_{ijkl}\epsilon_{kl}$$

The matrix C shown below and in Table 4.5 is a contracted form of the stiffness tensor C_{ijkl} , see Bower [2010, p. 78]. The indices first indicate the row, then the column.

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ & c_{22} & c_{23} & c_{24} & c_{25} & c_{26} \\ & & c_{33} & c_{34} & c_{35} & c_{36} \\ & & & c_{44} & c_{45} & c_{46} \\ & & & & c_{55} & c_{56} \\ & \text{sym.} & & & & c_{66} \end{bmatrix}$$

The matrix indices use Voight notation, see Barbero [2008, p. 310]. This contraction is possible because symmetries in the tensor reduce the number of independent material constants from 81 to 21.

Note that the indices in the stress and strain vectors are not the same as those used in the C matrix! The two indices in the stress and strain vectors indicate respectively the

plane whereupon the stress/strain operates and the direction in which said stress/strain points. In this case, 1 is x, 2 is y and 3 is z.

Table 4.4: properties of quasi isotropic laminate

generated by lamprop version 2023.10.10

Laminate stacking				
Layer	Weight [g/m ²]	Angle [°]	vf [%]	Fiber type
1	100	0	50	Hyer's carbon fiber
2	100	90	50	Hyer's carbon fiber
3	100	45	50	Hyer's carbon fiber
4	100	-45	50	Hyer's carbon fiber
5	100	-45	50	Hyer's carbon fiber
6	100	45	50	Hyer's carbon fiber
7	100	90	50	Hyer's carbon fiber
8	100	0	50	Hyer's carbon fiber

Physical properties		
Property	Value	Dimension
v_f	50	%
w_f	61.5	%
thickness	0.909	mm
density	1.43	g/cm ³
weight	1300	g/m ²
resin	500	g/m ²

Engineering properties					
In-plane			3D stiffness tensor		
E_x	48664	MPa	E_x	48693	MPa
E_y	48664	MPa	E_y	48693	MPa
E_z	15109	MPa	E_z	18206	MPa
G_{xy}	18388	MPa	G_{xy}	18405	MPa
G_{xz}	4239	MPa	G_{xz}	5088	MPa
G_{yz}	4241	MPa	G_{yz}	5088	MPa
ν_{xy}	0.3232	-	ν_{xy}	0.3228	-
ν_{yx}	0.3232	-	ν_{xz}	0.3124	-
α_x	3.05e-06	K ⁻¹	ν_{yz}	0.3124	-
α_y	3.05e-06	K ⁻¹			

Table 4.5: properties of quasi isotropic laminate (continued)

In-plane stiffness (ABD) matrix

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \begin{vmatrix} 4.9401 \times 10^4 & 1.5969 \times 10^4 & 0 & 0 & 0 & 0 \\ 1.5969 \times 10^4 & 4.9401 \times 10^4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.6716 \times 10^4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4.9487 \times 10^3 & 4.7560 \times 10^2 & 1.5370 \times 10^2 \\ 0 & 0 & 0 & 4.7560 \times 10^2 & 3.1042 \times 10^3 & 1.5370 \times 10^2 \\ 0 & 0 & 0 & 1.5370 \times 10^2 & 1.5370 \times 10^2 & 5.2709 \times 10^2 \end{vmatrix} \times \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \\ \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{Bmatrix}$$

Transverse stiffness (H) matrix

$$\begin{Bmatrix} V_y \\ V_x \end{Bmatrix} = \begin{vmatrix} 3.8552 \times 10^3 & 0 \\ 0 & 3.8536 \times 10^3 \end{vmatrix} \times \begin{Bmatrix} \gamma_{yz} \\ \gamma_{xz} \end{Bmatrix}$$

3D stiffness tensor (C), contracted notation

$$\begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{Bmatrix} = \begin{vmatrix} 5.8701 \times 10^4 & 2.1891 \times 10^4 & 9.4131 \times 10^3 & 0 & 0 & 0 \\ 2.1891 \times 10^4 & 5.8701 \times 10^4 & 9.4131 \times 10^3 & 0 & 0 & 0 \\ 9.4131 \times 10^3 & 9.4131 \times 10^3 & 2.0405 \times 10^4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5.0878 \times 10^3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5.0878 \times 10^3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.8405 \times 10^4 \end{vmatrix} \times \begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ 2 \cdot \epsilon_{23} \\ 2 \cdot \epsilon_{13} \\ 2 \cdot \epsilon_{12} \end{Bmatrix}$$

5 Tips and tricks

The 0° direction is generally in the length of the part or in the direction of the largest load.

The following section should be considered a *general guideline*. Sometimes there can be good reason to deviate from it.

5.1 Keep your laminates symmetric and balanced

Looking at the stacking of the layers, it should be symmetric w.r.t. the middle of the stack. So the following laminate is symmetric:

1. 0°
2. 45°
3. 90°
4. -45°
5. -45°
6. 90°
7. 45°
8. 0°

This is often shortened to “[0/45/90/-45]s”. The area weights of the layers should also be symmetric.

A balanced laminate is a laminate where for every layer at an angle on n° there is also a layer at $-n^\circ$. It is often added that for every 0° layer there should also be an equally sized 90° layer, but the author disagrees. For beam-like parts it is often desirable to have the majority of the fibers in the 0° direction.

Classical laminate theory strictly speaking is only valid for stacks of unidirectional layers. For woven fabrics and random oriented fiber products approximations are used.

5.2 Representing woven fabrics

A woven fabric is approximated as a $[0^\circ/90^\circ]$ s stack, where the weight of each layer is 1/4 of the total weight of the woven fabric. If warp and weft of the weave are not of equal weight, you should adjust the layers accordingly. Symmetry is important because a lone $[0^\circ/90^\circ]$ would exhibit tension/bending coupling that doesn't occur in a woven fabric. If the woven fabric is a small part of a larger stacking, you can use $[0^\circ/90^\circ]$ to represent a weave.

5.3 Representing non-wovens

Things like chopped strand mat (“CSM”), continuous filament mat (“CFM”) or other non-wovens can be approximated as a $[0^\circ/60^\circ/-60^\circ]$ s stack with the area weight evenly divided over the directions. Do keep in mind that the fiber volume fraction for such materials is significantly lower than for unidirectional or woven materials. For CSM it is unlikely to exceed 25% and for CFM 10–15% are typical values.

5.4 Align your fibers with the expected load

This is a no-brainer for tensile loads, but there is a twist. To counter torsion and shear loads, there should be layers of fibers in the $\pm 45^\circ$ direction. For bending loads the 0° layers should be at the outside of the part.

5.5 Laminate strength

As mentioned before, this program cannot predict the strength of laminates from the properties of the fibers and resin used in the layers; it is outside the scope of classical laminate theory.

Even stronger, the author does not believe that a general theory of laminate strength based on constituent properties is feasible due to the many different possible failure modes and the factors outside of the fiber and resin properties that influence the laminate. Examples of the latter are the void content, the degree of cure of the resin and

errors in cutting or placing the fibers. These are determined by type of production process used and the craftsmanship of the people involved.

However, the following guidelines have served the author well over the years.

For unidirectional layers loaded in the fiber direction, the strain at which either the fibers or the matrix fail in tension multiplied by the laminate's Young's modulus is the maximum allowed tensile stress.

The allowed compression stress for such layers is deemed to be 50%–60% of the allowed tensile stress

The strength of unidirectional layers in the $\pm 45^\circ$ or 90° directions is estimated as 10% of the strength in the 0° direction. This is the 10%-rule according to [Hart-Smith \[1992\]](#).

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Colofon

This document has been set with the “TeX Live”¹ implementation of the T_EX² typesetting software, using the L^AT_EX³ macros and specifically the MEMOIR⁴ style.

The main font used for the text is Alegreya⁵. The TeX Gyre Heros⁶ font is used for sans-serif text, while TeX Gyre Cursor is used for program names and program output.

¹<https://www.tug.org/texlive/>

²<http://nl.wikipedia.org/wiki/TeX>

³<http://nl.wikipedia.org/wiki/LaTeX>

⁴<http://www.ctan.org/tex-archive/macros/latex/contrib/memoir/>

⁵<https://github.com/huertatipografica/Alegreya>

⁶<http://www.gust.org.pl/projects/e-foundry/tex-gyre>